EXPERIMENT 05

Building Reports, Dashboards, and Custom Pages (UI Builder) in Zoho Creator

LOW CODE APPLICATION • LABORATORY REPORT

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Aim

To build and evaluate reports, dashboard-style summaries, and a custom Page Builder interface for a Gym Enrollment application in Zoho Creator.

Application overview

The app captures member identity, age, membership type, contact details, residential address, emergency contact, fitness notes, membership plan, preferred activities, and scheduling information. The existing application includes an enrollment form, two record views, and form workflows.

Evidence note

This report uses the supplied screenshots as evidence. The report views and workflows are visible in the screenshots; a completed dashboard or custom page is not shown, so that part is presented as the next implementation step rather than claimed as completed.

1. Existing Form and Data Model

The form is the data-entry layer of the application. It defines the fields users must complete and determines the structure of each stored enrollment record. The form-builder screenshot shows sections titled Member Details, Contact Details, Membership & Fitness Preferences, and Scheduling & Health Declaration. This grouping separates identity, communication, preferences, and declaration-related information, making a long form easier to scan.

The field palette includes Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, Drop Down, Radio, Multi Select, and Checkbox. The live form uses a Name field for the member, a Number field for age, a Radio field for membership type, contact fields, address details, and membership-related selections. Field labels shown to the user can differ from internal link names used by Deluge scripts; the link name must be checked before writing or debugging a script.

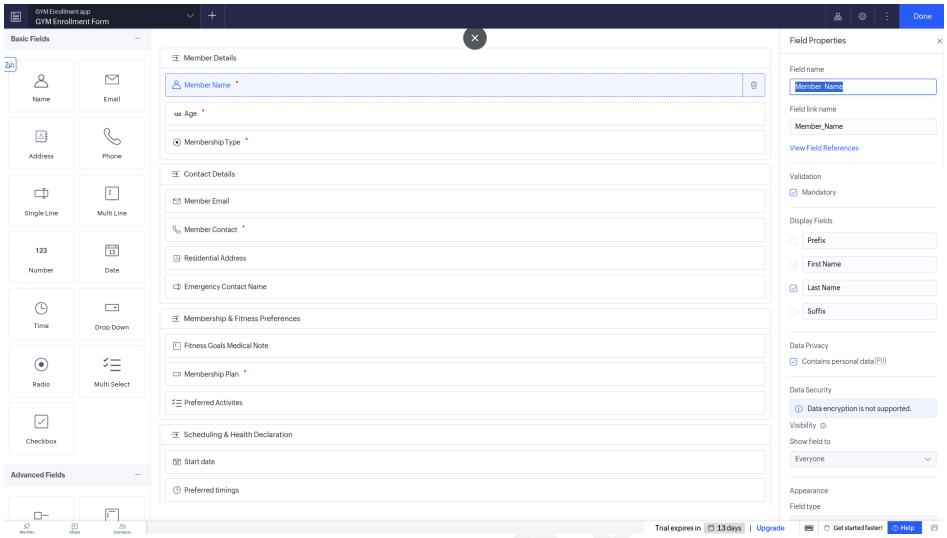


Figure 1. Form builder with section layout and configured fields.

Data-model considerations

Use a Number field for age so that comparisons are numeric rather than text-based. Use a Date field for the start date, an Email field for email validation, and a Phone field for contact details. A multi-select field suits several preferred activities, while a single-choice membership type is naturally represented by radio options or a dropdown. Required fields should be limited to information necessary for enrollment; unnecessary collection increases form friction and privacy risk.

The Member Name field is marked as containing personal data in the builder screenshot. Fitness or medical notes also deserve restricted access. Data design therefore affects both usability and privacy, not just storage.

2. Reports and Record Views

The application navigation contains “All Gym Enrollments” and “GYM enrollment report.” The first screenshot shows five records in the captured view. The second report view shows two records and includes Membership Plan as a column. These views demonstrate that submitted form data can be inspected in a structured table rather than opening each record individually.

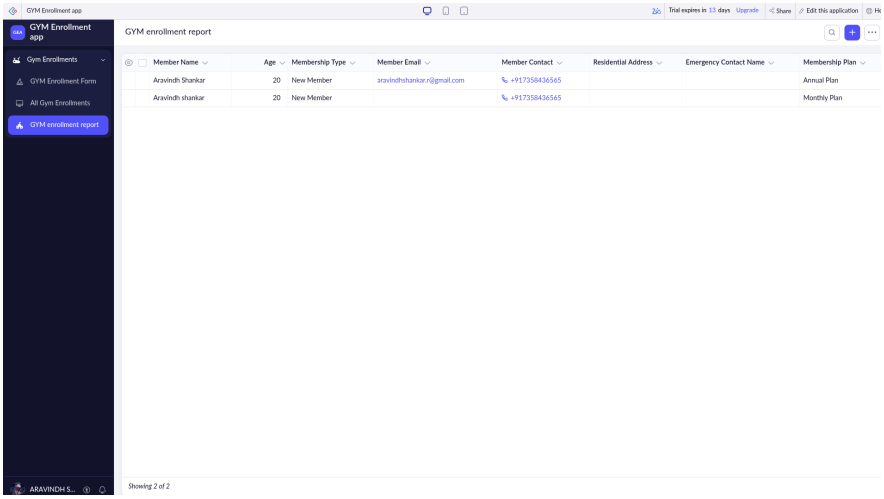


Figure 2. GYM enrollment report with membership details.

Report design and operational use

Reports support searching, sorting, filtering, and reviewing records. For a gym, front-desk staff may use the report to find a member, check the selected plan, verify contact details, or review the requested joining date. A useful report should display the fields needed for the task without exposing every stored field to every user.

Field	Why it is useful
Member Name	Identify the enrollment record.
Age	Review submitted value and detect invalid test data.
Membership Type	Distinguish new member, renewal, and corporate enrollment.
Email / Contact	Contact the member when follow-up is required.
Membership Plan	Compare or filter plan selections.
Start Date	Review planned joining dates, if included.

The student confirmed that filtering is working. Report filters should be tested against known records to ensure that the result set changes correctly and does not hide valid records unexpectedly. Report access must also follow the role permissions defined for the application.

3. Workflow Evidence and Automation

The Workflow screen shows three enabled form workflows: “Validate Start Date,” “Age check,” and “email confirmation.” This indicates that validation and notification logic has been configured. A separate Gmail screenshot shows a received message with the subject “Slot confirmed for the gym membership!” and a message confirming that the enrollment was received. This is evidence that the email action has executed successfully at least once.

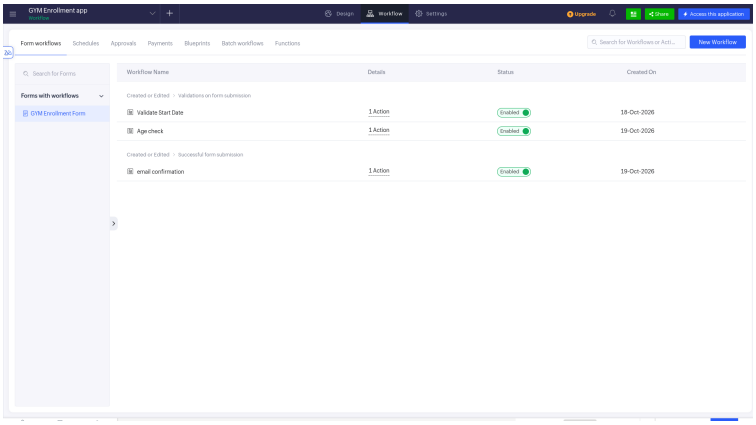


Figure 3. Enabled workflows configured for GYM Enrollment Form.

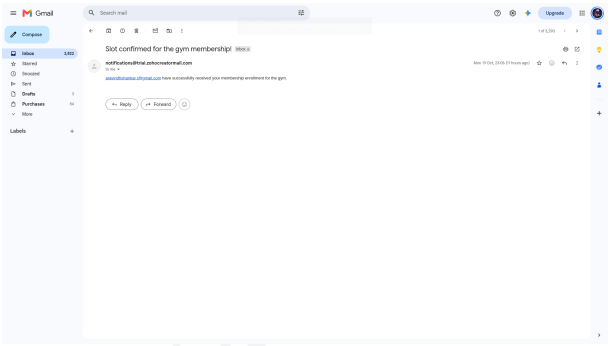


Figure 4. Confirmation email received in Gmail.

How the workflow stages should be separated

Validation should run at form submission and cancel the submission when a rule is violated. Confirmation email should run only after a valid record is saved. This separation prevents an invalid submission from being treated as a confirmed enrollment. Workflow names and enabled status alone do not prove that every rule works; the behaviour must be tested using valid and invalid inputs.

4. Validation Testing and Debugging

The supplied record-list screenshot contains several records with age 11, although the intended age range is 14–75. This is a concrete data-quality issue: the workflow is enabled, but the visible records show that the age rule has not consistently blocked invalid submissions. The cause may be an incorrect field link name, an incorrect condition, a missing cancellation statement, the wrong workflow event, or records created before the validation was fixed.

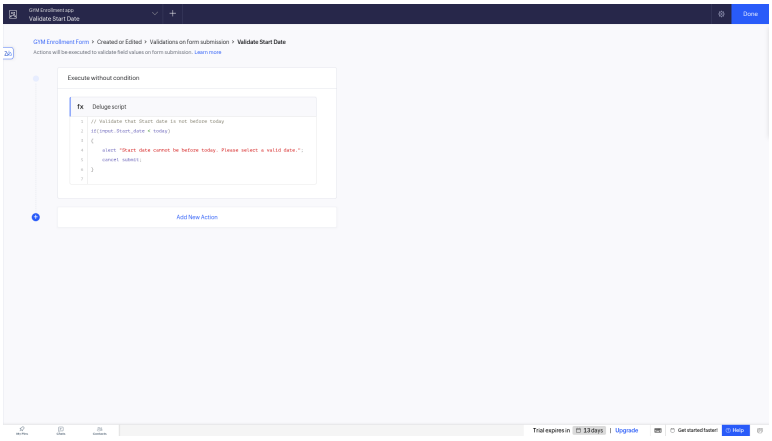


Figure 5. Deluge date validation script; the configured condition rejects dates earlier than today.

Boundary-value test plan

Test input	Expected outcome
Age 13	Submission blocked with clear message.
Age 14	Accepted if all other fields are valid.
Age 75	Accepted if all other fields are valid.
Age 76	Submission blocked with clear message.
Start date yesterday	Submission blocked.
Start date today / future	Accepted if other conditions pass.

Debugging sequence

Open the Age check workflow, confirm that it is a form-submission validation, verify the exact Age field link name, and check that the condition rejects values below 14 OR above 75. Confirm that the script cancels submission after displaying the message. Save the workflow and test the live form with all other required fields valid. Use a fresh record for each test and confirm whether it appears in the report.

Existing age-11 rows should be treated as invalid test data. Fixing the workflow will not automatically correct records that were already saved.

5. Dashboard and Custom Page Design

A dashboard combines key metrics and visual summaries so that a user can understand the state of the application without manually scanning every row. The supplied screenshots do not show a finished dashboard or custom UI Builder page. The design below is therefore the recommended next build step for completing this part of the experiment.

Component	Content	Purpose
Total Enrollments	Count of enrollment records	Quick view of volume.
Plan Distribution	Chart grouped by Membership Plan	Compare plan selections.
Recent Enrollments	Linked or embedded report	Inspect recent records.
Quick Actions	Open form / open report	Reduce navigation steps.
Validation Watch	Count or filter invalid test records	Support data-quality review during testing.

Suggested Page Builder procedure

1. Open the application editor and enter the Design area.
2. Create a page named Gym Membership Overview, using a blank layout if available.
3. Add a clear title and a metric or panel for total enrollments.
4. Add a chart grouped by Membership Plan and use record count as the value.
5. Add a link or embedded view of the enrollment report.
6. Arrange the components, save, and test the page in the live application.

Keep the page focused. One metric, one chart, and access to the detailed report are sufficient for a lab-scale dashboard. Avoid adding sensitive medical notes to a broad dashboard. If the account does not offer a desired widget, document the limitation and use a supported report or chart instead.

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6. Test Results and Evaluation

The following table distinguishes what is visible in the supplied evidence from what still needs verification. This distinction is important in a laboratory report: a configured feature is not automatically a proven feature, and a successful email test does not prove that every workflow condition is correct.

Test area	Evidence / status	Evaluation
Form structure	Sectioned form visible in builder	Configured
Record views	Two views shown in navigation	Configured
Filtering	Student reports filtering works	Reported working
Workflows	Three workflows shown as enabled	Configured; logic still needs testing
Email confirmation	Received email screenshot	Successful at least once
Age restriction	Age 11 records visible	Failure observed; fix and retest
Dashboard / custom page	No screenshot supplied	Not yet evidenced

Testing method

For each test, record a test ID, input values, expected result, actual result, and pass/fail status. Change only one variable at a time. For validation tests, keep other mandatory fields valid so that the reason for rejection is unambiguous. For reports, verify that the displayed count and filters match the saved records. For email, check the recipient, subject, message body, and whether the email is sent only after a successful save.

Data privacy and access

The app stores names, phone numbers, addresses, and fitness-related notes. Use test data when taking screenshots for submission, restrict reports to appropriate users, and avoid placing unnecessary personal information in the dashboard or confirmation email. Review sharing settings before making the app accessible beyond the lab team.

Acceptance criteria

The experiment can be considered complete when the form and reports work, filters return the intended records, age and date boundaries are correctly enforced, confirmation email is delivered after a valid submission, and a dashboard/custom page is either implemented and tested or clearly documented as unavailable in the current account.

7. Conclusion and References

The Gym Enrollment application demonstrates the main building blocks of a low-code business application: a structured form, stored records, report views, workflow automation, and email confirmation. The report views support day-to-day review of member information, while the workflow layer can enforce business rules and reduce manual notification work.

The most important finding is that the age restriction needs further debugging because records with age 11 are visible despite the intended 14–75 range. The presence of an enabled workflow is not enough to establish correctness. Boundary-value testing and confirmation that invalid records are not saved are required. The next UI task is to implement a focused dashboard or custom page that summarizes enrollment counts, plan distribution, and access to detailed records.

Learning outcomes

- Explained how reports organize stored form submissions for operational use.
- Identified the difference between record reports and a dashboard summary.
- Documented workflow and email evidence from the actual application.
- Recognized a validation defect from stored records and designed boundary tests.
- Outlined a practical Page Builder design with privacy and usability considerations.

References

1. Zoho Creator Help — User guides: https://www.zoho.com/creator/help/user-guides.html

2. Zoho Creator Help — Pages and Page Builder: https://help.zoho.com/portal/en/kb/creator/developer-guide/pages/create-and-manage-pages/articles/understand-pages

3. Zoho Creator Help — Workflows: https://www.zoho.com/creator/help/workflows/

4. Student-supplied screenshots of the Gym Enrollment form, reports, workflow list, Deluge script, and confirmation email (20 October 2026)

End of Experiment 05 Report

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